

Module 3: Software Architecture, Cohesion & UML

Modularity, 7 Cohesion Types, 6 Coupling Types, SOLID, UML

Module III: Software Architecture, Cohesion & UML — Topics Covered

1. Architectural Concepts & SOLID Principles

2. Modularity: Cohesion (7 Levels) & Coupling (6 Levels)

3. Architectural Styles (Layered, Microservices, MVC)

4. UML Structural: Use Case & Class Diagrams

5. UML Behavioral: Sequence, Activity & Statecharts

1. Cohesion and Coupling Quality Spectrum

Desirable Design Goal: High Cohesion within modules and Low (Loose) Coupling between modules.

Cohesion Spectrum (Low → High / Desirable)	Coupling Spectrum (High / Undesirable → Low / Desirable)
1. Coincidental (Worst)	1. Content Coupling (Worst - direct memory modification)
2. Logical	2. Common Coupling (Shared global variables)
3. Temporal	3. External Coupling
4. Procedural	4. Control Coupling
5. Communicational	5. Stamp Coupling (Passing composite data structure)
6. Sequential	6. Data Coupling (Best - passing scalar parameters)
7. Functional Cohesion (Best - performs single focused task)	